

SentinelAI

- **Problem Statement ID – 1**
- **Problem Statement Title-** AI-Powered Autonomous Smart Building Optimization Challenge
- **Theme-** Artificial Intelligence / Smart Buildings
- **PS Category-** Software
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- **Student ID:** 23BAI1265

SentinelAI: Autonomous AI-Powered Building Management System

Proposed Solution

How SentinelAI Works

1. Collects live telemetry from EnergyPlus
2. Builds a compact building-state context
3. Agent Council (LLM) optimizes:
 - Energy Consumption
 - Thermal Comfort
 - Carbon Emissions
 - Equipment Health
4. Safety Validator verifies all AI decisions
5. MCP Tool Server executes validated control actions
6. Dashboard visualizes system performance and Digital Twin

Innovation

1. Autonomous closed-loop AI control using EnergyPlus.
2. Open-Source LLM-powered Agent Council.
3. Model Context Protocol(MCP) tool execution.
4. Safety Validator with retry and Last-Known-Good fallback.
5. Rolling Context Builder for efficient LLM prompting.
6. Human-in-the-Loop approval for critical actions. Interactive Dashboard with Digital Twin visualization.
7. Baseline vs AI performance comparison with measurable energy savings

TECHNICAL APPROACH

Technologies Used

Simulation & AI

- EnergyPlus
- PyEnergyPlus API
- Open-Source LLM (via Ollama API)
- Model Context Protocol (MCP)

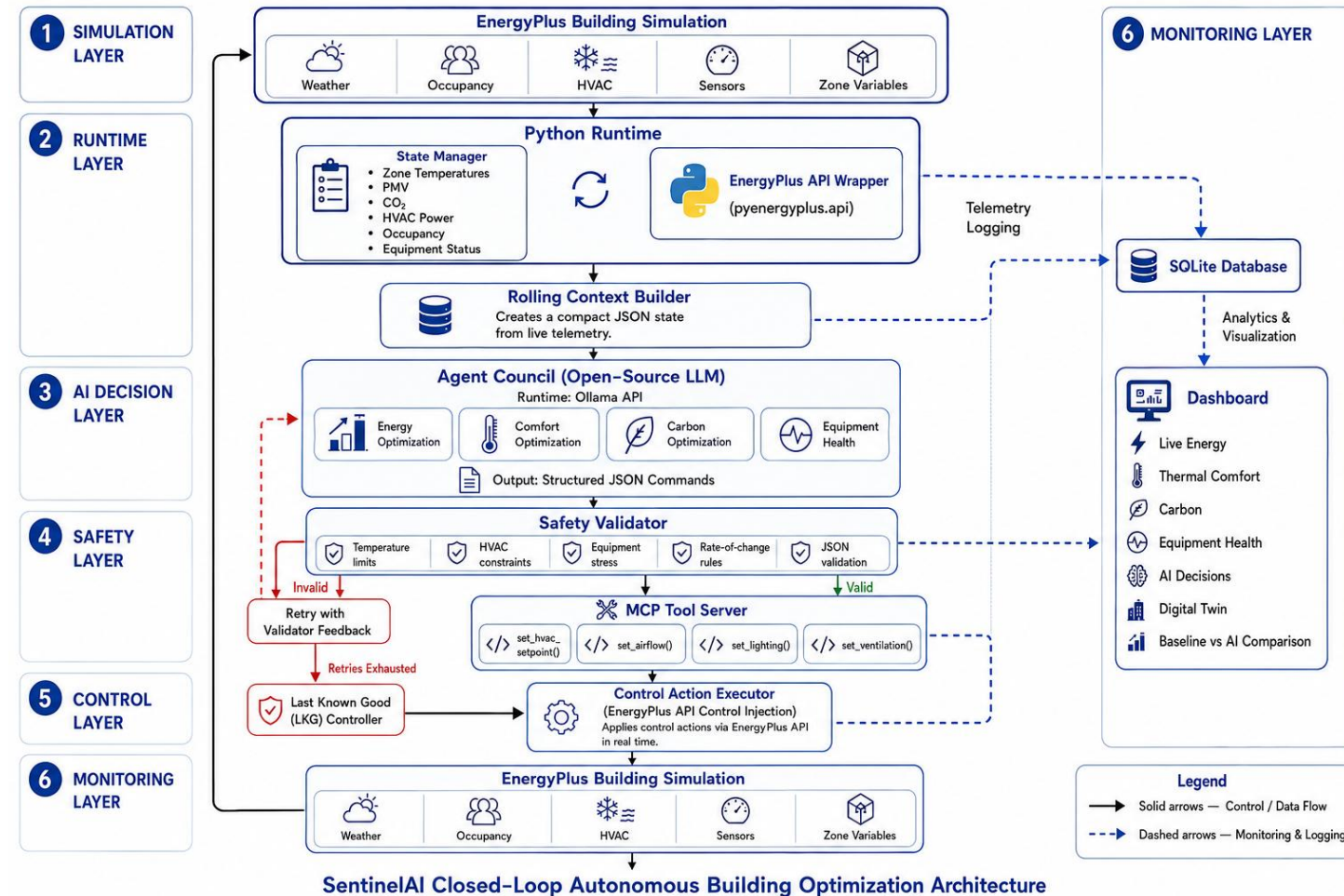
Backend

- Python
- FastAPI
- SQLite

Frontend

- HTML/CSS/JavaScript
- Plotly
- SVG Digital Twin

Architecture



FEASIBILITY AND VIABILITY

Feasibility

- **Industry-Standard Foundation:** Built on EnergyPlus, Python, FastAPI, and open-source LLMs.
- **Real-Time Closed-Loop Control:** Continuously monitors building telemetry and applies validated control actions.
- **Modular & Scalable:** Components (AI engine, dashboard, control layer) can be deployed independently and extended to larger buildings.
- **Cost-Effective:** Uses open-source technologies, reducing deployment and licensing costs.

Potential Challenges and Risks

- AI-generated unsafe control actions.
- Large EnergyPlus simulation logs.
- LLM response latency.
- Balancing energy efficiency with occupant comfort

Strategies

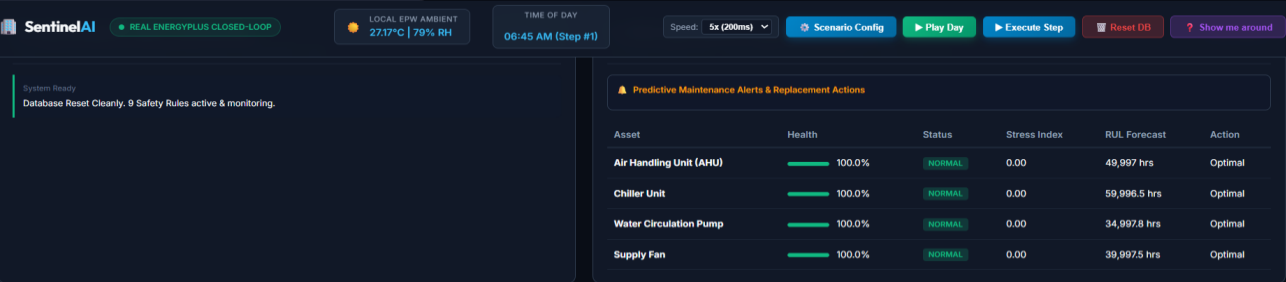
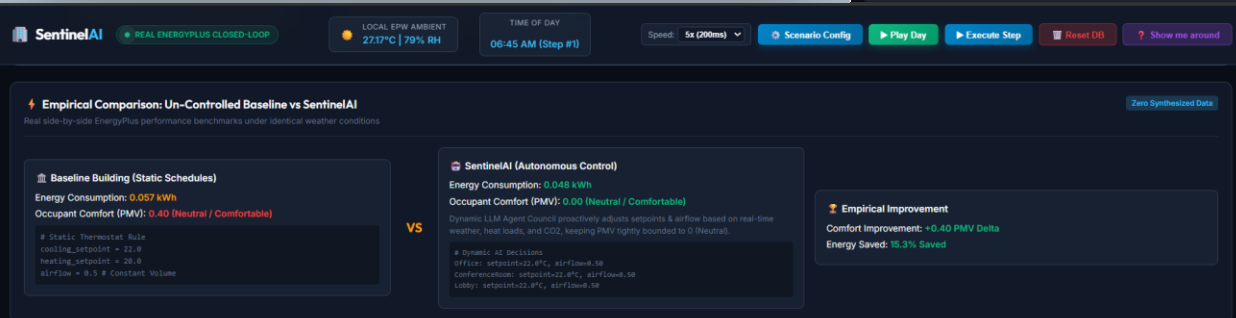
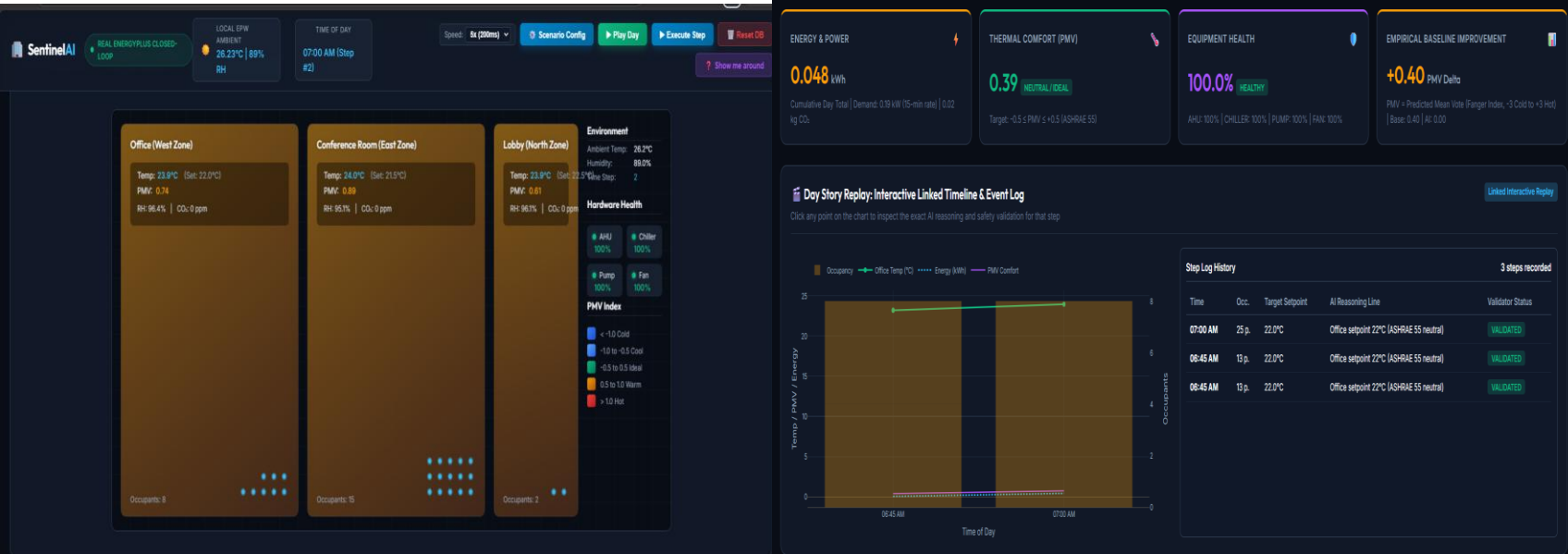
- **Safety Validator** checks every AI-generated control action before execution.
- **Retry & Last-Known-Good Fallback** ensures reliable operation if AI output is invalid.
- **Rolling Context Builder** summarizes telemetry, reducing prompt size and improving response time.
- **MCP-based modular architecture** enables integration with existing Building Management Systems and simplifies future expansion.

ARTIFACTS

GITHUB LINK: <https://github.com/APEXPRESS123207/SentinelAI>

VIDEO LINK: [Google drive](#) (Video also in github)

```
def get_default_rules() -> List[SafetyRule]:
    """Returns the standard set of safety rules for SentinelAI."""
    return [
        # Physical bounds
        SetpointRangeRule(min_c=19.0, max_c=26.0),
        AirflowRangeRule(min_flow=0.1, max_flow=1.0),
        LightingRangeRule(),
        # Rate Limits
        SetpointRateOfChangeRule(max_change_per_step=2.5),
        # Comfort
        PMVComfortRule(pmv_lower=-0.5, pmv_upper=0.5),
        CO2VentilationRule(co2_threshold_ppm=1000.0, min_airflow_high_co2=0.4),
        # Equipment protection
        ChillerMinRuntimeRule(),
        FanStaticPressureRule(fan_health_threshold=70.0, max_airflow_degraded=0.85),
        PumpHealthRule(critical_health_threshold=50.0),
    ]
```



RESEARCH AND REFERENCES

- Richard J. de Dear, Gail S. Brager, Thermal comfort in naturally ventilated buildings: revisions to ASHRAE Standard 55, Energy and Buildings, Volume 34, Issue 6, 2002, Pages 549-561, ISSN 0378-7788, [https://doi.org/10.1016/S0378-7788\(02\)00005-1](https://doi.org/10.1016/S0378-7788(02)00005-1).
(<https://www.sciencedirect.com/science/article/pii/S0378778802000051>)
- **ANSI/ASHRAE Standard 55-2023**
Thermal Environmental Conditions for Human Occupancy. (<https://www.ashrae.org/technical-resources/bookstore/standard-55-thermal-environmental-conditions-for-human-occupancy>)
- **U.S. Department of Energy – EnergyPlus Documentation**
<https://energyplus.net/documentation>